

Roadmap: fire impact curves for forest carbon

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At a glance

Goal. Find out how much of a forest's carbon is lost when it burns, based on how tall the trees are and how bad the fire is.

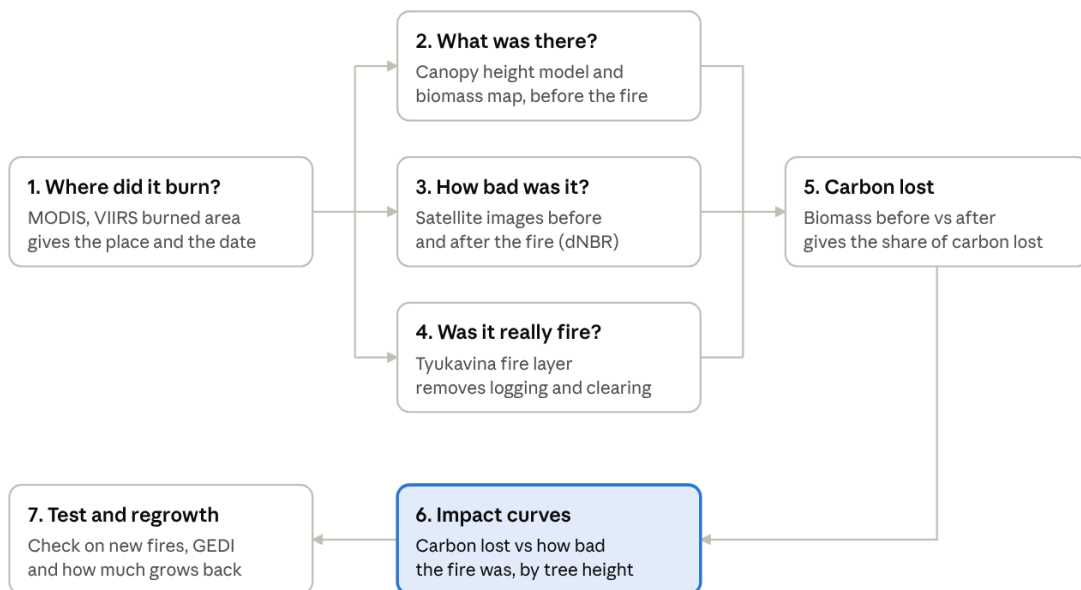
Scope. 4 weeks, one pilot region, fires from 2016 to 2020, open data only. We use 2016 to 2020 because biomass maps exist for every year in that period, so we always have a before and an after.

The idea in plain words

1. A **canopy height model** tells us how tall the trees were before the fire. It is a single snapshot, so on its own it cannot show damage. It tells us what was at risk.
2. **MODIS and VIIRS** satellites tell us where and when each fire happened.
3. **Satellite images from just before and just after** each fire tell us how badly each spot burned.
4. **Biomass maps from before and after** tell us how much carbon was really lost.
5. Put together, we get the answer we want. Trees of height X, hit by a fire of severity S, lose Y% of their carbon.

Seven simple questions turn fire and forest data into impact curves

Questions 2 to 4 are answered side by side for every fire found in question 1



roadmap · 7 questions and the data behind each

What the result looks like. The numbers below are made up, only to show the shape of the answer.

Tree height before the fire	Mild fire	Severe fire
Under 10 m	30% of carbon lost	90% lost
10 to 20 m	15% lost	75% lost
Over 20 m	5% lost	60% lost

With these curves we can take any project, read its tree height from the canopy model and estimate how much carbon a fire would remove.

Step 1. Where and when did it burn?

Why. We need a list of real forest fires, each with its outline on the map and its date.

Data

- **MODIS burned area (MCD64A1).** Marks every 500 m square that burned and the day it burned, from 2000.
- **VIIRS burned area (VNP64A1).** The same thing from a newer satellite, from 2012. Use both, as MODIS is close to the end of its life.
- **Global Fire Atlas.** Groups those burned squares into single fires, with start date, size and length. Saves us doing it by hand.
- **FIRMS.** Points where satellites saw active fire, with how much heat it gave off (FRP). We use this later in Step 3.
- **MTBS,** only if the region is in the US. Hand-checked fire outlines, good for checking our results.

Steps

1. Download the burned area maps for the pilot region, 2016 to 2020.
2. Take the list of fires from the Global Fire Atlas.
3. Keep only fires bigger than about 500 ha where at least half the area was forest.
4. Leave out the outer 500 m of each fire, where the 500 m maps are least accurate.

Result. A map layer of fires, each with an ID, outline, start and end date, and area.

Step 2. What was there before the fire?

Why. To measure damage we first need to know what was standing. This is where the canopy height model comes in.

Data

- **Canopy height model (GLAD, 2015).** Tree height for every 30 m square in 2015, so before all our fires. We use it to sort trees into height groups, not to measure loss. The newer ETH and Meta/WRI maps (around 2020) are more detailed, but they were made after most of these fires, so they would show the burnt forest. We keep them for later, to read tree height on Artio's projects.
- **Biomass map (ESA CCI Biomass).** How much wood, and so carbon, there is per hectare, every 100 m. Available for most years from 2015, so we can pick the year just before each fire.

- **Tree cover map (Hansen / GLAD).** Share of each 30 m square covered by trees in 2000, and the year the trees were lost, if they were.
- **Forest type map** (such as ESA WorldCover or a national map), to tell apart conifer, broadleaf and plantation.

Steps

1. Put all the maps on the same grid, so each square lines up across maps.
2. Cut them to the fire outlines from Step 1.
3. For each square, write down the biomass in the year before the fire. This is the carbon before.
4. Give each square a height group (under 10 m, 10 to 20 m, over 20 m) and a forest type.
5. Remove squares where trees were already gone before the fire.

Result. A table with one row per square, holding the fire ID, carbon before, tree height group and forest type.

Step 3. How bad was each fire?

Why. This gives the fire strength for our curves. A light ground fire and a fire that burns the tree tops do very different damage.

Data

- **Landsat 8 and 9.** Free satellite images every 8 days at 30 m.
- **Sentinel-2.** Free satellite images every 5 days at 20 m, to fill gaps when Landsat has cloud.
- **FIRMS fire heat (FRP)** from Step 1, as a second measure of fire strength.

Steps

1. For each fire, pick a clear image from the season before the fire, and one from the same season the year after. Using the same season avoids mistaking normal seasonal change for damage.
2. On both images, compute the burn index NBR. It compares two light bands (near-infrared and shortwave-infrared). Healthy trees score high, burnt ground scores low.
3. Subtract after from before. The difference is **dnBR**, and the bigger it is, the worse the fire.
4. Check one fire by eye (or against MTBS in the US) before running all fires.
5. Add the FIRMS fire heat for each square as a second measure.

As a rough guide, dnBR below 0.1 means unburnt, 0.1 to 0.27 a mild fire, 0.27 to 0.66 a moderate fire and above 0.66 a severe fire.

We use NBR and not NDVI. NDVI only measures how green things are, so it misses damage under thick canopy and bounces back as soon as grass regrows. NDVI is still useful as a quick cross-check.

Result. dnBR and fire heat added to the table for each square.

Step 4. Was it really fire?

Why. Inside a burnt area, trees can also be lost to logging, land clearing or storms. If we count those as fire damage, the curves will show more damage than fire really causes.

Data

- **Tyukavina fire layer.** For every 30 m square where Hansen shows tree loss, it says whether the cause was fire or something else. The published version covers 2001 to 2019. Check with GLAD whether a newer one covers 2020.
- **Hansen loss year** from Step 2, which says in which year the trees went.

Steps

1. Count tree loss as fire damage only if the Tyukavina layer says fire and the loss happened in the fire year or the year after.
2. Mark loss that comes 2 or more years later. It may be trees dying slowly from the fire, or logging of burnt trees. Keep it in a separate column.
3. Remove squares where the loss was not caused by fire.

Result. Each square is marked as fire loss, late loss or no loss.

Step 5. How much carbon was lost?

Why. This is the damage in our curves. It is measured as the share of carbon lost, from 0 (nothing lost) to 1 (everything lost).

Data. The biomass maps (ESA CCI Biomass) from before and after each fire, plus the marks from Step 4.

Steps

1. For each square, take the biomass from the year before the fire and from 1 year after. Do it again for 2 years after, as some trees die slowly.
2. Work out the share lost as (carbon before minus carbon after) divided by carbon before. Keep it between 0 and 1.
3. Do this for every burnt square, not only those where Hansen shows loss. Hansen only sees trees that are fully gone, so it misses partial damage from mild and moderate fires, and that is most of the curve.
4. Work at 100 m squares, the size of the biomass map.
5. Remove squares where the biomass map is too unsure to trust the change.

Result. The main table. One row per square, with fire strength (dNBR), tree height group, forest type and share of carbon lost after 1 and 2 years.

Step 6. Build the curves

Why. The curve turns the table into a simple rule. For a fire of a given strength, it says how much carbon trees of each height lose.

Steps

1. Split the table by tree height group and forest type.
2. In each group, sort the squares by fire strength (dNBR) into small steps of 0.05, and take the average share of carbon lost in each step.

3. Draw a smooth S-shaped curve through those points. Loss starts low for mild fires, rises fast, then levels off near total loss for severe fires.
4. Work out a range around each curve, to show how sure we are. Do it by fire, not by square, since squares in the same fire are alike.
5. Try the same with fire heat (FRP) instead of dNBR and see which predicts loss better.
6. Merge groups that have too few fires to give a stable curve.

Result. One curve per height group and forest type, with its range, and a chart of each curve over its data.

Step 7. Test the curves and track regrowth

Why. We need to know the curves work on fires they have not seen, and how much of the lost carbon comes back. Carbon that grows back matters a lot for reversal risk.

Data

- **GEDI.** A laser on the International Space Station that measures tree height and biomass in small spots, from 2019. Very accurate, but only in scattered spots.
- **Biomass maps and Landsat images** for each year after the fire, up to 2022.

Steps

1. At the start, put aside 1 in 5 fires. Do not use them to build the curves. At the end, check how well the curves predict their loss.
2. Find GEDI spots measured both before and after a fire, and compare the real loss with what the curve says.
3. For the 2016 and 2017 fires, track biomass each year up to 2022. Report how much of the lost carbon came back after 1, 3 and 5 years.
4. Split loss that stays lost from loss that grows back.

Result. How far off the curves are on unseen fires, and a regrowth curve per forest type.

What we have at the end

- ☐ Code for each step, in one Git repo, that runs from the fire list to the curves
- ☐ The main table from Step 5
- ☐ The curves per tree height and forest type, with their ranges, saved in a format CLIMADA can read
- ☐ Test and regrowth results from Step 7
- ☐ A 2-page note on how it was done, what it cannot do yet and next steps, plus a short presentation to the team

All work stays on internal systems. No Artio project data leaves the company.

STEPS

- 1. Select areas for fire and forests. It’s important that we have different biomes. **Forest type map** (such as ESA WorldCover or a national map), to tell apart conifer, broadleaf and plantation.
- 2. VIIRS to know if there were fires in those areas.
- 3. **Canopy height model (GLAD, 2015)**. Tree height for every 30 m square in 2015, so before all our fires. We use it to sort trees into height groups, not to measure loss. The newer ETH and **Meta/WRI maps (around 2020)** are more detailed, but they were made after most of these fires, so they would show the burnt forest. We keep them for later, to read tree height on Artio's projects.

Height of the tree. So if a fire happens 2024, you’d take the canopy height 2023. We’re interested in dense forest, places with a good number of trees.

- 4. **First we measure the biomass 2023.Biomass map (ESA CCI Biomass).**
- 5. **Biomass 2024.**
- 6. **Hansen loss year** from Step 2, which says in which year the trees went.

VIIRS intensity of fire.
Sentinel-2 (GEE) Landsat / **dNBR**

Thickness of the tree, age of the tree or more parameters then even better.

Tree height before the fire	You make the categories		Severe fire
Under 10 m	30% of carbon lost		90% lost
10 to 20 m	15% lost		75% lost
Over 20 m	5% lost		60% lost

% of biomass loss
100%

90%

10%

Fire intensity 200k 250k 400k 500k

Humidity and moisture.

The equation (Create an equation that fits the curve)